The additional semantics encoded by a knowledge graph are defined by an ontology—a formal specification of the concepts, relationships, and rules that exist within a particular domain.

This semantic foundation transforms a collection of data points into a coherent model that mirrors how the physical world actually works. Instead of writing custom code to handle every relationship, you define the domain once in an ontology, and the knowledge graph automatically provides rich, queryable context for all your data.

NOTE

The term *knowledge graph* was first popularized by Google around 2012 following it's acquisition of Freebase, itself an online collection of shared knowledge. Today the panel of facts related to a search term that you get when performing a Google search is retrieved from its knowledge graph which as of 2020 reportedly held 500 billion facts about 5 billion entities.

5.4.2 Building a knowledge graph as a labeled property graph

Labeled property graphs are a popular way to represent a knowledge graph where both nodes and edges are *labeled*, assigned a label—the same as a class in an ontology—and both nodes and edges can have multiple key value properties attached. A simple example of a knowledge graph implemented as a labeled property graph based on RealEstateCore is shown in figure 5.7. This example shows how you might model a thermostat located in a room on the second floor of a hotel. You can imagine for a hotel with several thousand rooms spread over many levels how much more complex this model becomes.